¶ case file 01 / 07 · agentic automl platform — filed 2025-09 · last verified 2026-07

status: in progress — core shipped, run ledger pending

you opened this file · jul 25, 2026

Agentic AutoML Platform

A private, GitHub-backed agentic AutoML platform. Datasets and domain documents become auditable pipeline decisions — and a human approval gate holds every generated action before it alters the workflow.

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role	capstone engineer — my slice below
timeframe	2025-09 to present
stack	typescript · react 19 · express 5 · postgresql · docker · langgraph · mcp
repo	private — yadava5/ai-augmented-auto-ml-toolchain
live demo	agentic-automl-platform.vercel.app ↗ (https://agentic-automl-platform.vercel.app)

private repository

evidence on file — private-safe

private proof: github evidence — the current github repository is yadava5/ai-augmented-auto-ml-toolchain and its readme identifies the product as agentic automl platform. the repository is private, so this case file shows private-safe evidence instead of a public source link.

[problem] · § as found

Between a raw dataset and a useful model sits a chain of repetitive judgment: ingestion, feature decisions, training, evaluation, deployment packaging. Automate the chain carelessly and the judgment disappears with the labor.

constraints —

- make pipeline decisions auditable instead of opaque.
- support domain documents through retrieval and mcp-based orchestration.
- require human approval before generated actions alter workflow state.
- keep training workflows reproducible with containerized execution.
- validate the product flow with browser-level checks.

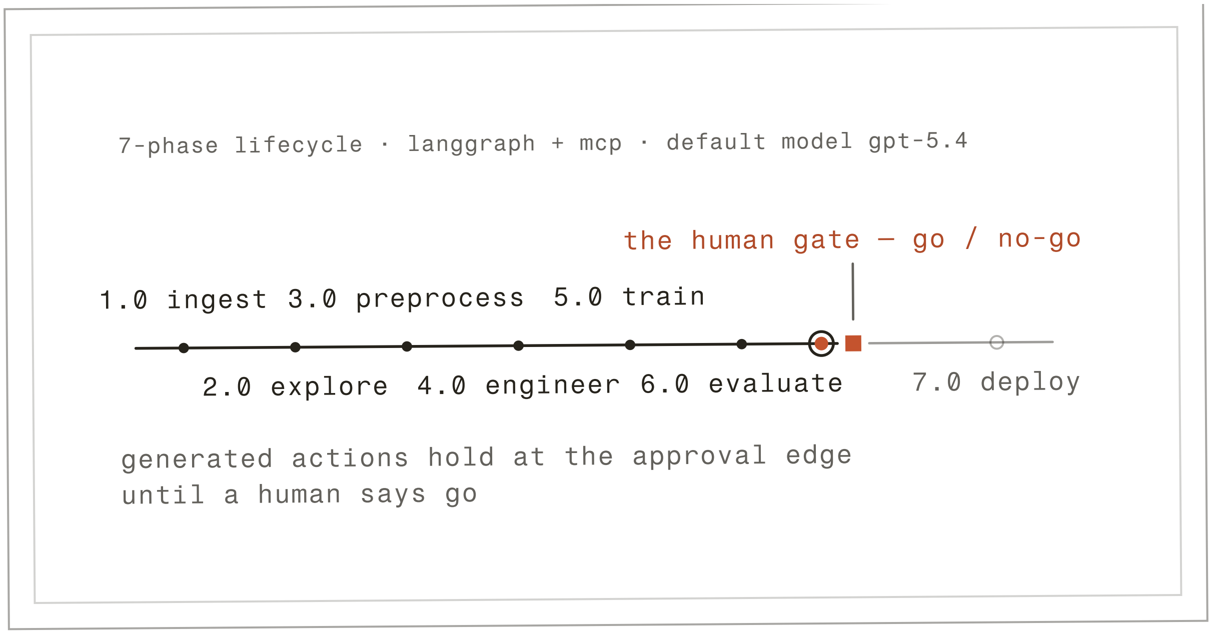

fig. 1 — the halt, echoed — the run crosses the 7-phase lifecycle and stops at the human gate.

description: A drawn figure that runs — not a screenshot. A card-scale echo of the pinned home scene: the ink run token crosses the lifecycle's first six phases and halts at the clay human gate — 7.0 deploy never resolves, because deploying is a person's decision, not this card's. The phases and the gate are the platform's real 7-phase, LangGraph + MCP-orchestrated workflow (default model GPT-5.4); run 041 stays awaiting its human.

[architecture] · § fig. 2, inked

A React and TypeScript interface coordinates Express/PostgreSQL services, LangGraph + MCP orchestration, notebook-based training workflows, Docker execution, and Playwright evaluation.

react 19 ui dataset and workflow surface	express 5 api pipeline orchestration
langgraph + mcp agentic workflow routing	docker runtime reproducible runs
postgresql 16 run metadata	playwright evals workflow validation

the circuit —

react 19 ui → express 5 api workflow requests

express 5 api → langgraph + mcp approved actions

express 5 api → docker runtime training jobs

docker runtime → postgresql 16 run records

playwright evals → react 19 ui browser proof ↺ closes the loop

generated actions hold at the approval edge
until a human says go

fig. 2 — the gated loop, inked. clay marks the gates: where a check can stop the run.

run · model	metrics	status
038 · log-reg	▒▒	approved
039 · random forest	▒▒	approved
040 · gbm	▒▒	approved
041 · xgboost	▒▒	awaiting approval

fig. 3 — experiment registry, transcribed private-safe excerpt.

metrics withheld — private repository; see the boundary rows below.

[decisions] · § as filed

d1 — use langgraph + mcp for orchestration · accepted

The platform needs phase-aware routing, tool calls, and auditable decisions tied to domain context. [t1]

> t1. tradeoff — more infrastructure than a simple model runner, but better for traceable workflows.

d2 — keep human approval gates · accepted

Generated preprocessing, training, and deployment actions should be reviewed before they alter the workflow. [t2]

> t2. tradeoff — approval gates slow down full automation, but they make the system safer and easier to debug.

d3 — containerize execution · accepted

Training runs need reproducible environments. [t3]

> t3. tradeoff — docker adds setup cost but reduces machine-specific drift.

> method slip — eval protocol
>
> protocol: not yet documented — see corrections.

validation

walk the 8 claims ⟶

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01 claim: The platform lives in the private repo yadava5/ai-augmented-auto-ml-toolchain; its README titles it Agentic AutoML Platform.

method: live github metadata check
date: 2026-07

artifact: no linkable artifact — described only
[private-safe]

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02 claim: Upload, EDA, NL-to-SQL, preprocessing, training, experiments, and deployment are the seven lifecycle phases.

method: read from the senior design poster and the local repo
date: 2026-05

artifact: see fig. 6 — the expo poster
[private-safe]

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03 claim: HPO, multi-model search, notebook training, and automated workflow evaluation are built into the platform.

method: poster + presenter deck, checked against the local repo
date: 2026-05

artifact: see fig. 5 — the presenter deck
[private-safe]

04 claim: Presenter slide 8 records the stack and validation posture: all-green tests, coverage, logs, packages, and migrations.

method: transcribed from the presenter artifact
date: 2026-05

artifact: see fig. 5 — the presenter deck
[private-safe]

05 claim: My slice of the build: the Monaco/Jupyter runtime with live WebSocket sync, Docker sandbox constraints, the eval runner, and the Optuna study streaming UI.

method: as presented — the presenter artifact names this work
date: 2026-05

artifact: see fig. 5 — the presenter deck
[private-safe]

06 claim: Training runs execute in a Dockerized runtime for reproducibility.

method: poster architecture panel + local repo audit
date: 2026-05

artifact: see fig. 6 — the expo poster
[private-safe]

outcomes

07 claim: A dataset and a goal become a structured, auditable workflow — planned and argued for by agents that still cannot press go.

method: the product's design, described — not an outcome metric
date: date not recorded

artifact: no linkable artifact — described only
[private-safe]

08 claim: Pipeline decisions run through LangGraph and MCP tool calls rather than free-form output.

method: poster workflow-state panel

date: 2026-05

artifact: see fig. 6 — the expo poster

[private-safe]

ci rows link the public run · repo pins are the exact commits verified 2026-07.

what i’m not claiming —

no per-run metrics are published here. the registry excerpt shows run, model, and status only — a demo-data run ledger with a complete metric trail has not shipped yet.

the repository is private, so source is not inspectable from this page. the evidence is the expo poster, the presenter deck, and the private-safe registry capture — verifiable in interview.

[corrections] · § the register

note · 2026-07

Per-run metrics remain withheld while the repository is private; the eval protocol for the platform's own model runs is not yet publicly documented. Nothing here has been retracted — this register is waiting on a demo-data run ledger.

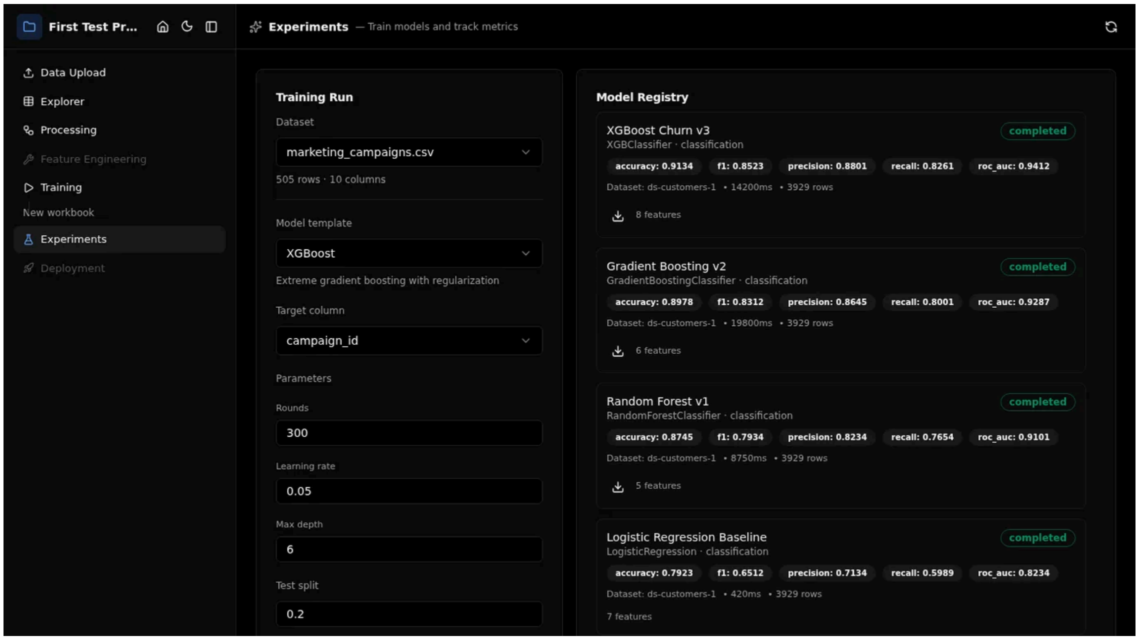


fig. 4 — private-safe experiment registry screenshot.

description: local automl repository, demo data · 2026-06 · open to inspect ⟶

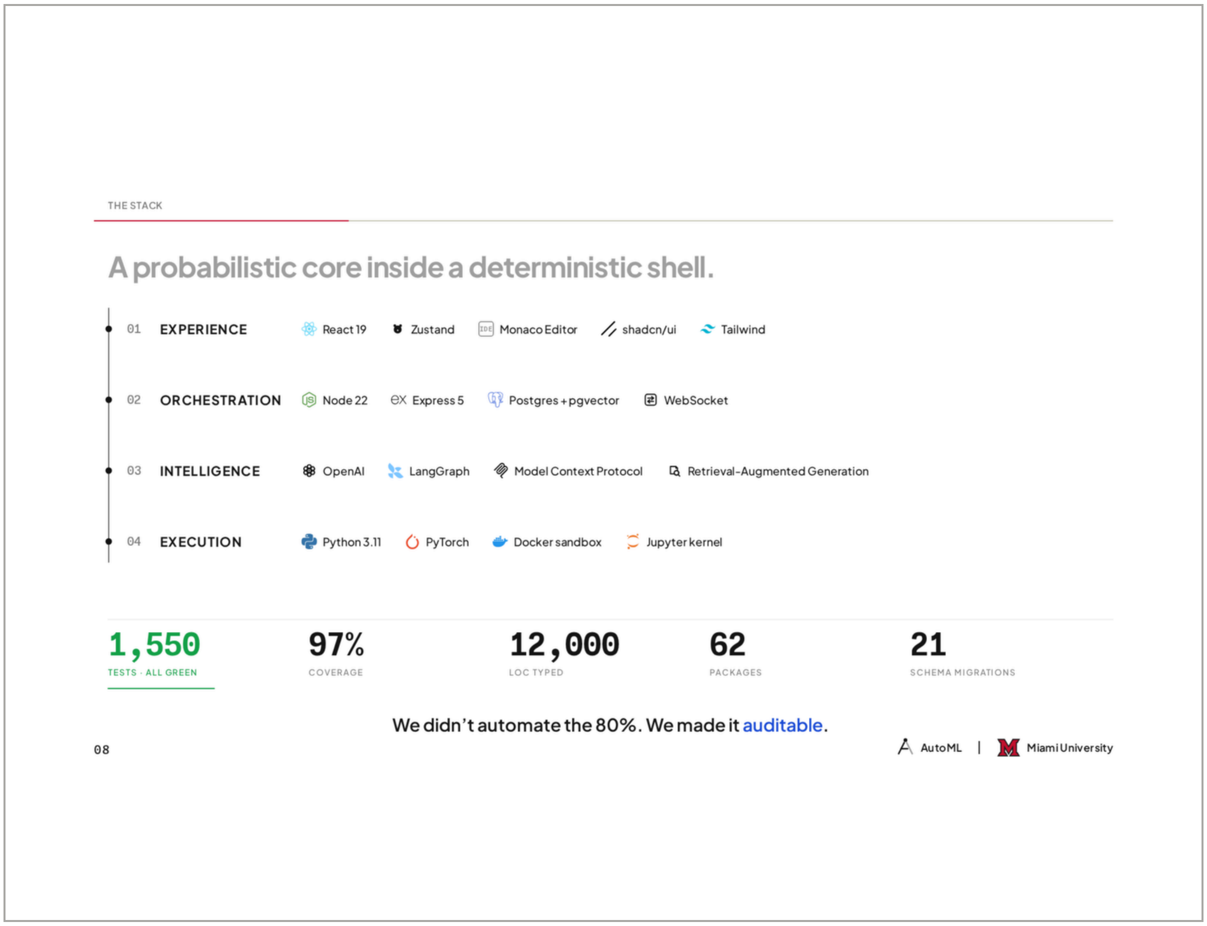


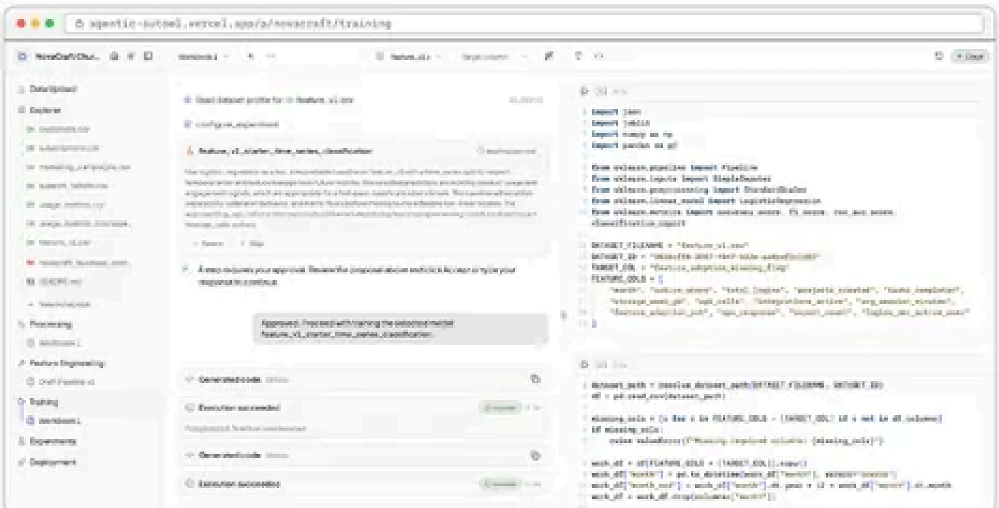
fig. 5 — presenter stack proof.

description: senior design presenter deck, slide 8 · 2026-05 · open to inspect ⟶

§2 | THE PRODUCT

An agent you can approve.

Each turn is a tool-call cluster the agent asks you to sign off on — *every design choice on the record*, reversible from a checkpoint.



34 TOOLS
Per-stage allowlist · every call explicitly approved

2 EDITORS
Monaco & Jupyter kernel · live WebSocket diff · reversible

6 LEDGER TABLES
Postgres audit log · every turn replayable

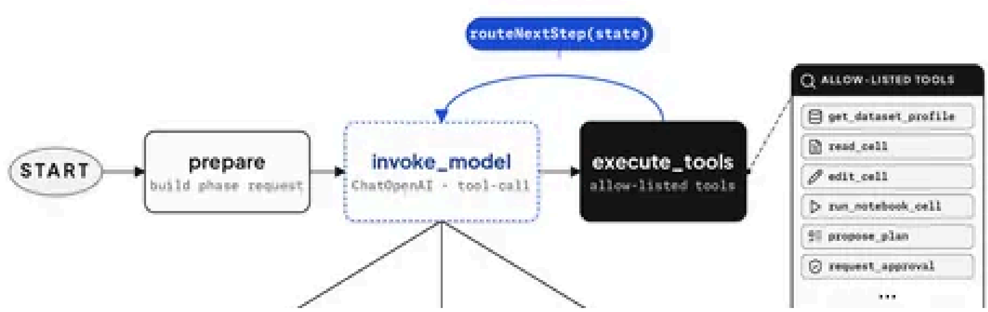
SCREENSHOT Training phase, captured live from agentic-automl.vercel.app via Playwright

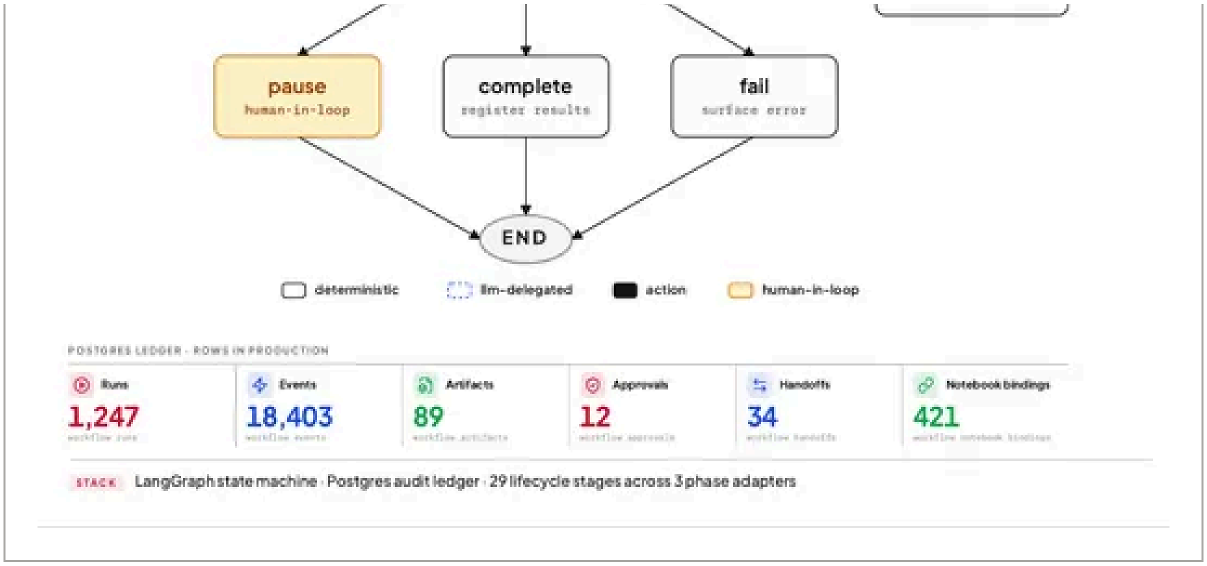
§2 — an agent you can approve

§3 | THE ARCHITECTURE

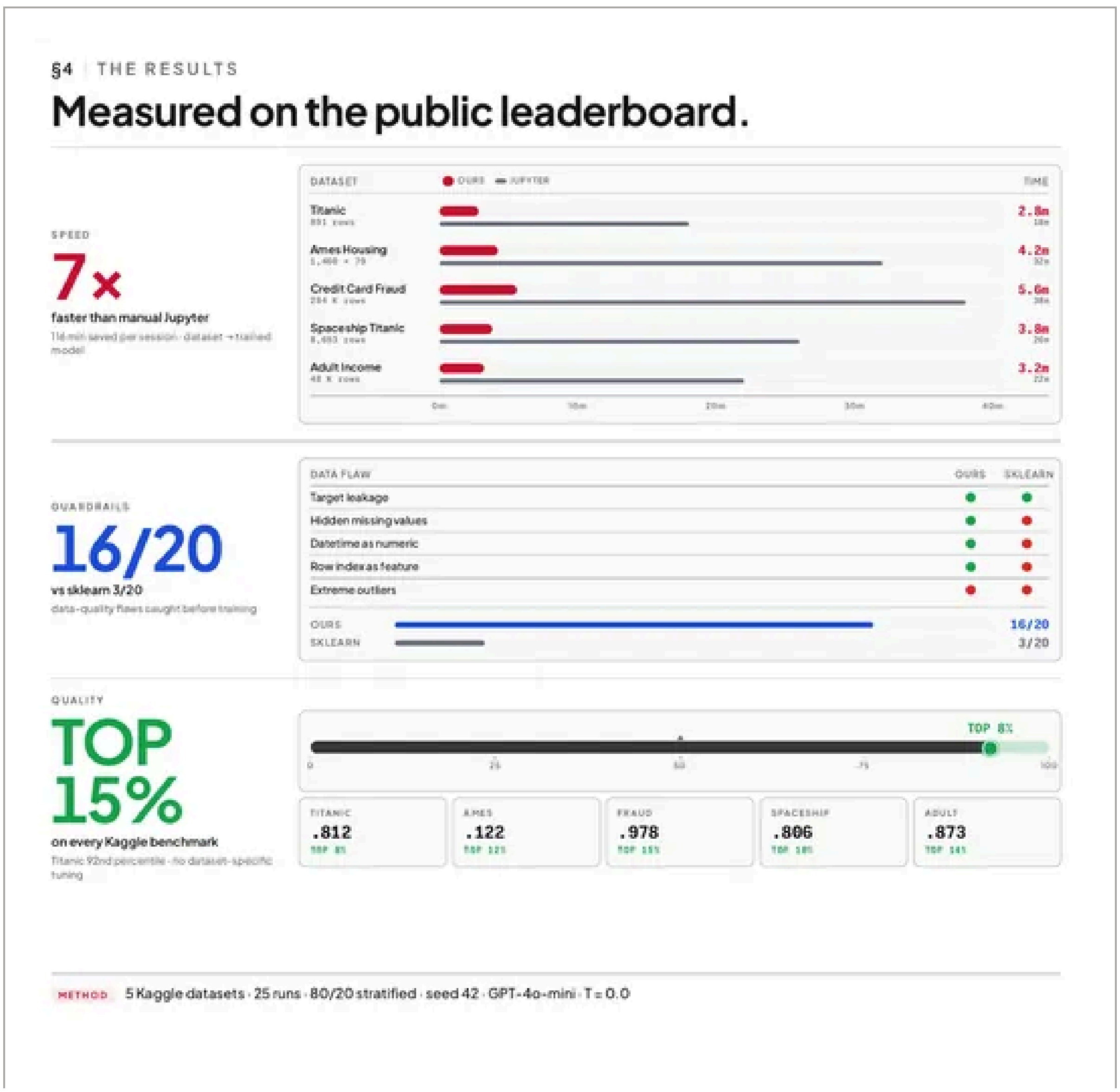
One LangGraph. Every step auditable.

A **probabilistic core** (`invoke_model`) wrapped in a **deterministic shell** — one shared state graph, three phase adapters, 29 lifecycle stages.





§3 — one langgraph, every step auditable



§4 — measured on the public leaderboard

fig. 6 — expo poster proof.

panels cropped at column width from the checked-in capture — titles quoted from the poster.

description: senior design expo poster, spring 2026 · 2026-05

open the full poster →

case file 01 / 07

back to the work ←

the evidence index →

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set in fraunces, newsreader & fragment mono

two inks. one line. seven chapters.